Fourier Analysis for CT Systems

- The Fourier Series
- The Fourier Transform

Fourier Analysis

- An alternative to time-domain convolution to describe the input-output operations of linear systems
- **Fourier techniques** transform time-domain waveform $x(t)$ to a frequency-domain representation $X(j\Omega)$
- **Fourier series**: Periodic functions can be represented as infinite sums of sines and cosines
- **Fourier Transform**: extends the fourier representation to non-periodic functions
- Laplace Transform further generalization of the Fourier transform
- Fourier analysis gives some insights to signal processing techniques such as filtering and synthesis

Definition

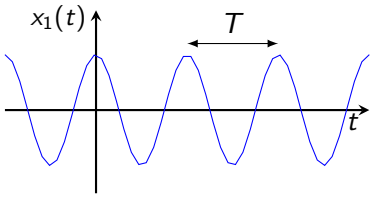
Periodic function: In general, a *periodic function* is a function that satisfies the relationship

$$x(t) = x(t + T)$$

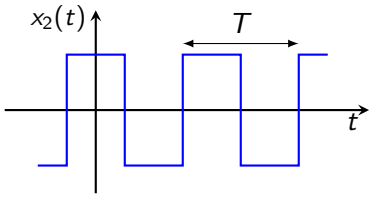
for all t, or $x(t) = x(t + nT)$, where $n = \pm 1, \pm 2, \pm 3, \ldots$ where T is defined as the *period*.

Examples of Periodic functions in time

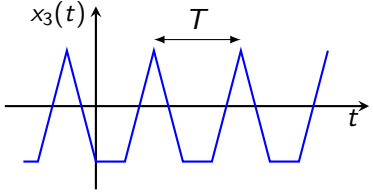
Cosine



Square wave



Triangular / Sawtooth



Fundamental frequency

Definition

The *fundamental angular frequency* Ω_0 (in radians/second) of a periodic waveform is defined directly from the period

$$\Omega_0 = \frac{2\pi}{T}$$

and the *fundamental frequency* F_0 (in Hz) is

$$F_0 = \frac{1}{T}$$

so that

$$\Omega_0 = 2\pi F_0$$

Harmonically related wave forms

Definition

Two wave forms whose periods, or frequencies, are related by a simple integer ratio are said to be *harmonically related*.

In general when harmonically related sinusoid are added together the resulting wave form is also periodic with a repetition period equal to the fundamental period.

Example 1

Consider two periodic signals:

$$x_1(t) \text{ with } T_1 = 12s$$

$$x_2(t) \text{ with } T_2 = 4s$$

If the fundamental frequency is define by $x_1(t)$, then $\Omega_0 = 2\pi/12$, and $x_2(t)$ has a frequency of $3\Omega_0$.

$x_1(t)$ and $x_2(t)$ are said to be harmonically related, and $x_2(t)$ have a frequency which is the *third* harmonic of the fundamental Ω_0.
If the wave forms are added:

$$g(t) = x_1(t) + x_2(t)$$

then $g(t)$ will repeat at intervals defined by the longest period of the two, every 12 seconds.

Example 2: Adding harmonically-related sinusoids

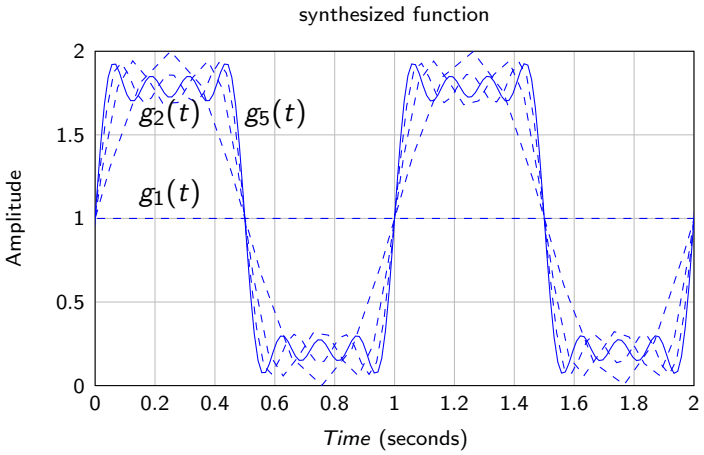
A family of wave forms is formed by adding the first N of up to five component functions, that is:

$$g_N(t) = \sum_{n=1}^{N} x_n(t) \qquad 1 < N \leq 5$$

where

$$\begin{aligned} x_1(t) &= 1 \\ x_2(t) &= \sin(2\pi t) \\ x_3(t) &= \frac{1}{3}\sin(6\pi t) \\ x_4(t) &= \frac{1}{5}\sin(10\pi t) \\ x_5(t) &= \frac{1}{7}\sin(14\pi t) \end{aligned}$$

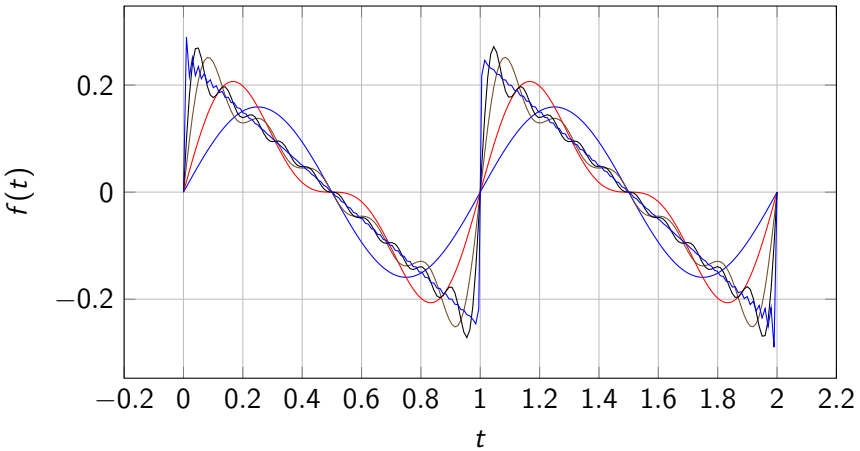
Sum of harmonically related sinusoids



Synthesis of a periodic waveform by a summation of harmonically related sinusoidal components. The period is $T = 1sec$, the fundamental period.

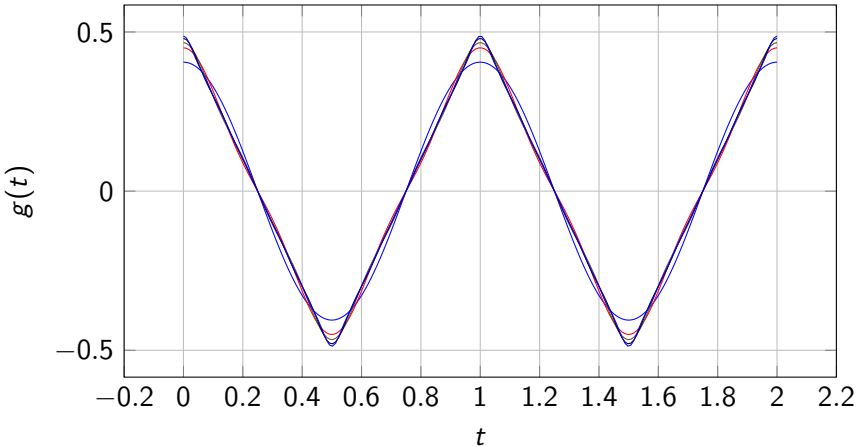
Other periodic wave forms: Sawtooth

$$f(t) = \sum_{n=1}^{\infty} \frac{1}{n} \sin(n\Omega_0 t)$$



Other periodic wave forms: Triangular

$$g(t) = \sum_{n=1, n=odd}^{\infty} (-1)^{\left(\frac{n-1}{2}\right)^2} \frac{1}{n^2} \sin(n\Omega_0 t)$$



Fourier Series: Harmonic components

The *Fourier series* representation of a real periodic function $x(t)$, with period T, is a sum of harmonically related sinusoidal components.

The nth harmonic component can be expressed as:

$$x_n(t) = a_n \cos(n\Omega_0 t) + b_n \sin(n\Omega_0 t) \tag{1}$$

where $n\Omega_0$ are the harmonic frequencies which are integer multiples of the fundamental Ω_0.

The nth harmonic component has frequency $n\Omega_0 = 2\pi n/T$.

Fourier Series: Harmonic components (second form)

Another representation of the sinusoidal harmonic component can be written as:

$$x_n(t) = A_n \sin(n\Omega_0 t + \phi_n) \tag{2}$$

where $n\Omega_0$ are the harmonic frequencies, and $n\Omega_0 = 2\pi n/T$.
Each component is expressed as a single sinusoid with amplitude A_n and phase ϕ_n

Fourier Series: Harmonic components

The harmonic components represented in (1) and (2) are directly related by

$$A_n \sin(n\Omega_0 t + \phi_n) = A_n \sin(\phi_n)\cos(n\Omega_0 t) + A_n \cos(\phi_n)\sin(n\Omega_0 t)$$

$$\begin{aligned} a_n &= A_n \sin(\phi_n) \\ b_n &= A_n \cos(\phi_n) \end{aligned} \tag{3}$$

and

$$\begin{aligned} A_n &= \sqrt{a_n^2 + b_n^2} \\ \phi_n &= \tan^{-1}(a_n/b_n) \end{aligned} \tag{4}$$

Fourier Series of CT signals

The *Fourier series* representation of an arbitrary periodic wave form $x(t)$, with period T, is an infinite sum of harmonically related sinusoidal components. The two equivalent forms are expressed as:

$$x(t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n \cos(n\Omega_0 t) + b_n \sin(n\Omega_0 t)) \tag{5}$$

$$= \frac{1}{2}a_0 + \sum_{n=1}^{\infty} A_n \sin(n\Omega_0 t + \phi_n) \tag{6}$$

Knowledge of the *fundamental frequency* Ω_0, and the set of **Fourier coefficients** $\{a_n\}$ and $\{b_n\}$ (or $\{A_n\}$ and $\{\phi_n\}$) are sufficient to completely define the wave form $x(t)$.

Fourier Series: Complex Exponential form

Each harmonic component $x_n(t)$ is expressed in terms of complex exponentials instead of real sinusoids using Eulers identity.

$$\begin{aligned} x_n(t) &= a_n \cos(n\Omega_0 t) + b_n \sin(n\Omega_0 t) \\ &= \frac{a_n}{2}\left(e^{jn\Omega_0 t} + e^{-jn\Omega_0 t}\right) + \frac{b_n}{2j}\left(e^{jn\Omega_0 t} - e^{-jn\Omega_0 t}\right) \\ &= \frac{1}{2}(a_n - jb_n)e^{jn\Omega_0 t} + \frac{1}{2}(a_n + jb_n)e^{-jn\Omega_0 t} \\ &= X_n e^{jn\Omega_0 t} + X_{-n} e^{-jn\Omega_0 t} \end{aligned} \tag{7}$$

where the new coefficients

$$\begin{aligned} X_n &= \frac{1}{2}(a_n - jb_n) \\ X_{-n} &= \frac{1}{2}(a_n + jb_n) \end{aligned} \tag{8}$$

are complex number and complex conjugates, that is $X_{-n} = X_n^*$.

Fourier Series: Complex Exponential Form

Based on harmonically related complex exponentials, the Fourier series is expressed in compact form.

$$x(t) = \sum_{n=-\infty}^{+\infty} X_n e^{jn\Omega_0 t} \tag{9}$$

The sum is taken over positive and negative values of n, where the coefficient of terms for positive and negative values of n are complex conjugates.

$$X_{-n} = X_n^* \tag{10}$$

so that knowledge of the coefficients X_n for $n \geq 0$ is sufficient to define the function $x(t)$.

Fourier Series: Other nomenclature

It is also common to use the index k to denote the kth Fourier coefficient (C_k), thus in other text:

$$x(t) = \sum_{k=-\infty}^{+\infty} C_k e^{jk\Omega_0 t} \quad \text{and} \quad C_k = \frac{1}{T}\int_{t_0}^{t_0+T} x(t)e^{-jk\Omega_0 t}dt$$

In other text the frequency ω_0 is used as

$$x(t) = \sum_{k=-\infty}^{+\infty} C_k e^{jk\omega_0 t} \quad \text{and} \quad C_k = \frac{1}{T}\int_{t_0}^{t_0+T} x(t)e^{-jk\omega_0 t}dt$$

Example 3

A periodic function $x(t)$ consists of five components

$$x(t) = 2 + 3\sin(100t) + 4\cos(100t) + 5\sin(200t + \pi/4) + 3\cos(400t)$$

It can be expressed as a finite complex Fourier series by expanding each term.

$$\begin{aligned} x(t) &= 2 + \frac{3}{2j}\left(e^{j100t} - e^{-j100t}\right) + \frac{4}{2}\left(e^{j100t} + e^{-j100t}\right) \\ &\quad + \frac{5}{2j}\left(e^{j(200t+\pi/4)} - e^{-j(200t+\pi/4)}\right) + \frac{3}{2}\left(e^{j400t} + e^{-j400t}\right) \\ &= 2 + \left(2 + \frac{3}{2j}\right)e^{j100t} + \left(2 - \frac{3}{2j}\right)e^{-j100t} \\ &\quad + \left(\frac{5}{2\sqrt{2}} + \frac{5}{2\sqrt{2}j}\right)e^{j200t} + \left(\frac{5}{2\sqrt{2}} - \frac{5}{2\sqrt{2}j}\right)e^{-j200t} \\ &\quad + \frac{3}{2}e^{j400t} + \frac{3}{2}e^{-j400t} \end{aligned}$$

Example (continued)

$$x(t) = 2 + \left(2 + \frac{3}{2j}\right)e^{j100t} + \left(2 - \frac{3}{2j}\right)e^{-j100t} + \left(\frac{5}{2\sqrt{2}} + \frac{5}{2\sqrt{2}j}\right)e^{j200t} + \left(\frac{5}{2\sqrt{2}} - \frac{5}{2\sqrt{2}j}\right)e^{-j200t} + \frac{3}{2}e^{j400t} + \frac{3}{2}e^{-j400t}$$

The fundamental frequency is $\Omega_0 = 100$ radians/second, the time-domain functions contains harmonics $n = 1, 2, 3,$ and 4. The complex Fourier coefficients are:

$$X_0 = 2$$

$$X_1 = 2 - \frac{3}{2}j \qquad X_{-1} = 2 + \frac{3}{2}j$$

$$X_2 = \frac{5}{2\sqrt{2}}(1 - 1j) \qquad X_{-2} = \frac{5}{2\sqrt{2}}(1 + 1j)$$

$$X_3 = 0 \qquad X_{-3} = 0$$

$$X_4 = \frac{3}{2} \qquad X_{-4} = \frac{3}{2}$$

Example (continued)

The finite Fourier series expressed in complex form using the coefficients as:

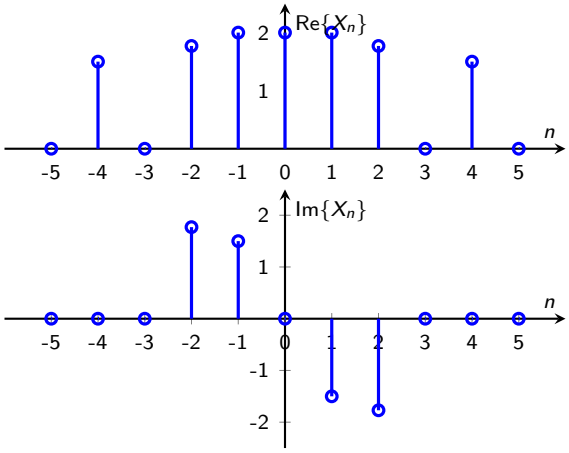
$$x(t) = \sum_{n=-4}^{4} X_n e^{jn100t}$$

Important notes:

- We adopt the nomenclature of using capital letters to represent the Fourier coefficients in the *complex* notation.
- The set of coefficients $\{G_n\}$ represent the function $g(t)$ and $\{Y_n\}$ represent the coefficients of $y(t)$
- Lower case coefficients $\{a_n\}$ and $\{b_n\}$ represent the real Fourier coefficients of any function of time.

Example (continued)

The plot (line spectra) of the real and imaginary parts of each harmonic component of $x(t)$ are shown below.



Computation of the Fourier Coefficients

For any periodic waveform $x(t)$ with period T, the Fourier coefficients are computed as:

$$a_n = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\cos(n\Omega_0 t)dt \tag{11}$$

$$b_n = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\sin(n\Omega_0 t)dt \tag{12}$$

The complex Fourier coefficients are computed as:

$$X_n = \frac{1}{T}\int_{t_0}^{t_0+T} x(t)e^{-jn\Omega_0 t}dt \tag{13}$$

where t_0 is any arbitrary time origin.

Fourier Analysis and Synthesis

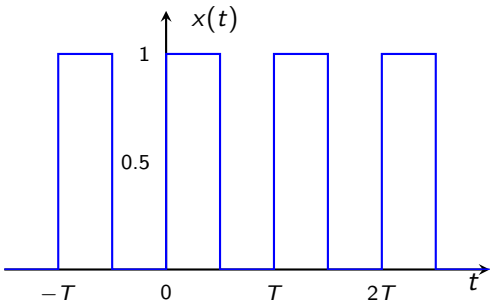
Table: Summary of analysis and synthesis equations for Fourier analysis and synthesis

	Sinusoidal formulation	Exponential formulation
Synthesis	$x(t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n \cos(n\Omega_0 t) + b_n \sin(n\Omega_0 t))$	$x(t) = \sum_{n=-\infty}^{+\infty} X_n e^{jn\Omega_0 t}$
Analysis	$a_n = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\cos(n\Omega_0 t)dt$ $b_n = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\sin(n\Omega_0 t)dt$	$X_n = \frac{1}{T}\int_{t_0}^{t_0+T} x(t)e^{-jn\Omega_0 t}dt$

Example 4

Find the complex and real Fourier series representation of the periodic square wave $x(t)$ with period T.

$$x(t) = \begin{cases} 1, & 0 \leq t < T/2, \\ 0, & T/2 \leq t < T \end{cases}$$



Example: Solution

Solving for X_0 separately:

$$X_0 = \frac{1}{T}\int_0^{T/2} e^{j0}dt = \frac{1}{2}$$

Solving for the other coefficients:

$$\begin{aligned} X_n &= \frac{1}{T}\int_0^{T/2}(1)e^{-jn\Omega_0 t}dt \\ &= \frac{-1}{jn\Omega_0 T}\left[e^{-jn\Omega_0 t}\right]_0^{T/2} \\ &= \frac{j}{2n\pi}\left[e^{-jn\pi} - 1\right] \end{aligned}$$

since $\Omega_0 T = 2\pi$. Because $e^{-jn\pi} = -1$ when n is odd, and $+1$ when n is even.

Example 4: Solution

We have:

$$X_n = \begin{cases} -j/n\pi, & \text{if n is odd;} \\ 0, & \text{if n is even;} \end{cases}$$

The square wave can be written as the complex Fourier series

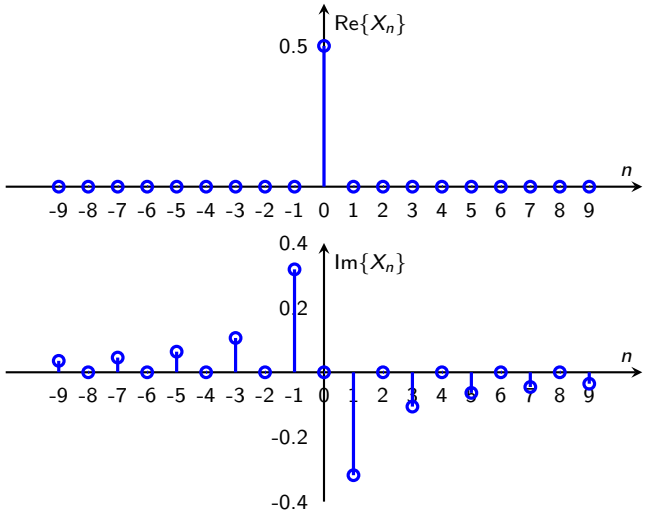
$$x(t) = \frac{1}{2} + \sum_{m=1}^{\infty} \frac{j}{(2m-1)\pi} \left(e^{-j(2m-1)\Omega_0 t} - e^{j(2m-1)\Omega_0 t} \right)$$

where the terms $X_n e^{j\Omega_n t}$ and $X_{-n} e^{-j\Omega_n t}$ have been combined in the summation. Expanding using the Euler formulas, the cosine terms cancel.

$$\begin{aligned} x(t) &= \frac{1}{2} + \sum_{m=1}^{\infty} \frac{j}{(2m-1)\pi} \left(e^{-j(2m-1)\Omega_0 t} - e^{j(2m-1)\Omega_0 t} \right) \\ &= \frac{1}{2} + \frac{2}{\pi} \left(\sin(\Omega_0 t) + \frac{1}{3}\sin(3\Omega_0 t) + \frac{1}{5}\sin(5\Omega_0 t) + \frac{1}{7}\sin(7\Omega_0 t) + \dots \right). \end{aligned}$$

Example 4: Solution

The line spectra of the square wave $x(t)$ is given below:



1. Existence of the Fourier Series

The Dirichlet conditions, guarantee the existence of the Fourier series for a given periodic waveform $x(t)$.

- The function $x(t)$ must be absolutely integrable over any period.

$$\int_{t_0}^{t_0+T} |x(t)|dt < \infty$$

  for any t_0.

- There must be at most a finite number of *maxima* and *minima* in the function $x(t)$ within any period
- There must be at most a finite number of discontinuities in the function $x(t)$ within any period, and all such discontinuities must be finite in magnitude.

The Dirichlet conditions are sufficient conditions, but not necessary.

2. Linearity of the Fourier Series representation

The Fourier analysis and synthesis operations are linear. Consider two functions $g(t)$ and $h(t)$ with the same periods T, their complex Fourier coefficients are:

$$G_n = \frac{1}{T}\int_{t_0}^{t_0+T} g(t)e^{-jn\Omega_0 t}dt$$

$$H_n = \frac{1}{T}\int_{t_0}^{t_0+T} h(t)e^{-jn\Omega_0 t}dt$$

a third function $x(t)$ defined as the weighted sum

$$x(t) = ag(t) + bh(t)$$

where a and b are constants. The linearity property states that the Fourier coefficients of $x(t)$ are:

$$X_n = aG_n + bH_n$$

3. Even and Odd Functions

If $x(t)$ is an even function: $x(t) = x(-t)$, then the complex Fourier series has coefficients X_n that are purely real. The real series only contains only cosine terms.

$$x(t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(n\Omega_0 t)$$

If $x(t)$ is an odd function: $x(t) = -x(-t)$, the coefficients X_n are purely imaginary. The real series contains only sine terms.

$$x(t) = \sum_{n=1}^{\infty} b_n \sin(n\Omega_0 t)$$

4. Fourier series of a time-shifted function

If the periodic function $x(t)$ has a Fourier series with complex coefficients X_n. The Fourier series of a time-shifted function $g(t) = x(t + \tau)$ is given by

$$G_n = e^{jn\Omega_0\tau} X_n$$

If the nth spectral component is written in terms of its magnitude and phase:

$$x_n(t) = A_n \sin(n\Omega_0 t + \phi_n)$$

then,

$$\begin{aligned} x_n(t+\tau) &= A_n \sin(n\Omega_0(t+\tau) + \phi_n) \\ &= A_n \sin(n\Omega_0 t + \phi_n + n\Omega_0\tau) \end{aligned}$$

The additional phase shift $n\Omega_0\tau$, due to the time shift τ, is proportional to the frequency of the component $n\Omega_0$.

5. Interpretation of the zero-frequency term

The coefficient X_0 and a_0 correspond to a harmonic component with zero frequency. Complex analysis gives:

$$X_0 = \frac{1}{T}\int_{t_1}^{t_1+T} x(t)dt$$

and the real analysis equation gives:

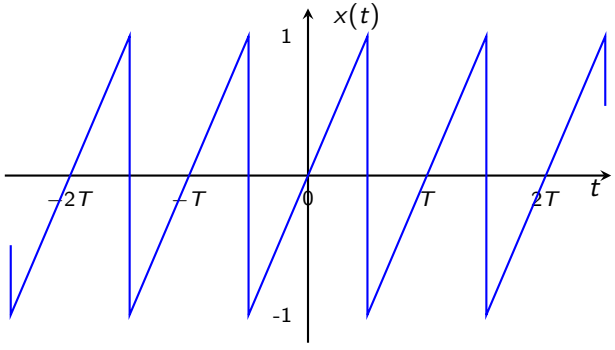
$$\frac{1}{2}a_0 = \frac{1}{T}\int_{t_1}^{t_1+T} x(t)dt$$

The average value of the function over one period. Adding a constant to a function only changes the value of X_0 or a_0

Example 5

Find the Fourier series representation for the periodic "saw-tooth" waveform with period T

$$x(t) = \frac{2}{T}t, \quad |t| < T/2$$



Example 5; Solution

The complex coefficients are

$$X_n = \frac{1}{T}\int_{-T/2}^{T/2} \frac{2}{T} t e^{-jn\Omega_0 t} dt$$

Integrating by parts:

$$\begin{aligned} X_n &= \frac{2j}{n\Omega_0 T^2}\left[t e^{-jn\Omega_0 t}\Big|_{-T/2}^{T/2} + \frac{2}{T^2}\int_{-T/2}^{T/2} \frac{1}{jn\Omega_0} e^{-jn\Omega_0 t} dt \right. \\ &= \frac{j}{2n\pi}\left[e^{-jn\pi} + e^{jn\pi} \right] + 0 \\ &= \frac{j}{n\pi}\cos(n\pi) \\ &= \frac{j(-1)^n}{n\pi} \qquad n \neq 0, \quad \text{since } \cos(n\pi) = (-1)^n \end{aligned}$$

Example 5: Solution

The zero frequency coefficient must be evaluated separately

$$X_0 = \frac{1}{T}\int_{-T/2}^{T/2}\left(\frac{2}{T}t\right)dt = 0$$

The Fourier series is

$$\begin{aligned} x(t) &= \sum_{n=1}^{\infty}\frac{j(-1)^n}{n\pi}\left(e^{jn\Omega_0 t} - e^{-jn\Omega_0 t}\right) \\ &= \sum_{n=1}^{\infty}\frac{2(-1)^{n+1}}{n\pi}\sin(n\Omega_0 t) \\ &= \frac{2}{\pi}\left(\sin(\Omega_0 t) - \frac{1}{2}\sin(2\Omega_0 t) + \frac{1}{3}\sin(3\Omega_0 t) - \frac{1}{4}\sin(4\Omega_0 t) + \ldots\right) \end{aligned}$$

Fourier Analysis for Aperiodic or Transient Signals

- Many waveforms found in practice are not periodic, many system excitations are transient or aperiodic. They are limited in time, occur only once and decays to zero over time.

- The Fourier analysis of transient phenomena is based on the limiting behavior of the Fourier series of a periodic function $x_p(t)$ as the period T approach infinity.

Fourier Series of a Rectangular pulse

Consider the Fourier series of a rectangular pulse with fixed width Δ and its period T is varied.

$$x(t) = \begin{cases} 1, & |t| < \Delta/2 \\ 0, & \Delta/2 \leq |t| \leq T - \Delta/2. \end{cases}$$

$$\begin{aligned} X_n &= \frac{1}{T} \int_{-\Delta/2}^{\Delta/2} e^{-jn\Omega_0 t} dt \\ &= \frac{j}{2n\pi} \left[e^{-jn\Omega_0 \Delta/2} - e^{jn\Omega_0 \Delta/2} \right] \\ &= \frac{1}{n\pi} \sin(n\pi\Delta/T) \\ &= \frac{\Delta}{T} \frac{\sin(n\pi\Delta/T)}{(n\pi\Delta/T)} \qquad n \neq 0, \end{aligned}$$

Fourier Series of a Rectangular pulse

Solving for X_0:

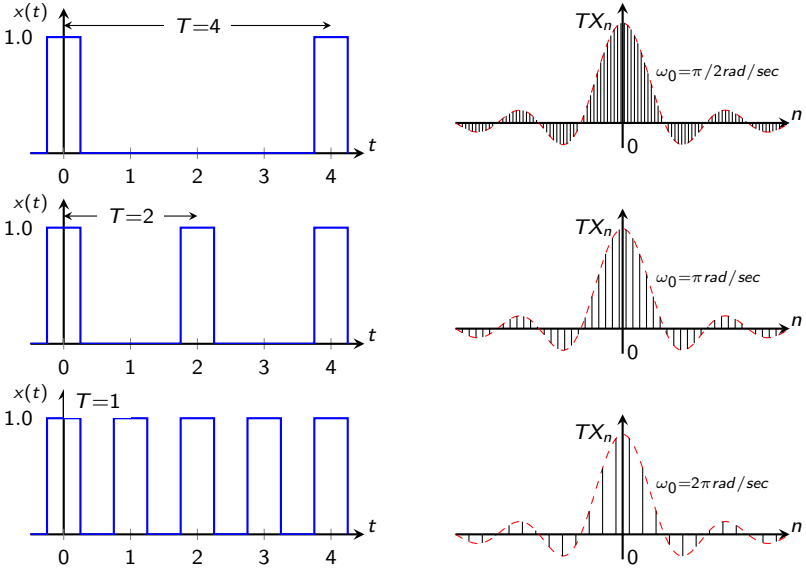
$$X_0 = \frac{1}{T}\int_{-\Delta/2}^{\Delta/2}(1)dt = \frac{\Delta}{T}$$

And we have:

$$X_n = \frac{\Delta}{T}\frac{\sin(n\pi\Delta/T)}{(n\pi\Delta/T)}$$

The spectral lines are placed at intervals $\Omega_0 = 2\pi/T$ rad/sec.

Line spectra of a Rectangular pulse



Fourier Series of a Rectangular pulse

To investigate as T is varied, we define a continuous function $X(\Omega)$.

$$X(\Omega) = \frac{\sin(\Omega\Delta/2)}{\Omega\Delta/2}$$

The Fourier coefficients can be computed directly from $X(\Omega)$

$$\begin{aligned} X_n &= \frac{\Delta}{T}\frac{sin(\Omega\Delta/2)}{\Omega\Delta/2} \\ &= \frac{\Delta}{T}X(\Omega)|_{\Omega=n\Omega_0} \end{aligned}$$

The function $X(\Omega)$ is an *envelop* function of the line spectra, depends only on the pulse width Δ and is independent of the period T (except for the scaling factor).

Fourier Series of a Rectangular pulse

General observation on the behavior of X_n as T vareis

- As T increases, Ω_0 decreases: spacing between adjacent lines in the spectrum decreases.
- As T increases, scaling factor Δ/T decreases, magnitude of the spectral lines diminish as T approach infinity
- The "shape" of the spectrum is defined by $X(\Omega)$ and is independent of T.

Fourier Transform

It can be shown that for an aperiodic function $x(t)$ that is non-zero in a defined interval Δ, with its periodic extension $x_p(t)$.

The limit of the Fourier series representation of $x_p(t)$ as the period T approach infinity leads to the formulation of the Fourier transform of $x(t)$ and is defined as:

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt \tag{14}$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega) e^{j\Omega t} d\Omega \tag{15}$$

Eq. (14) is the *forward* transform, $X(j\Omega)$ is a continuous function of frequency. Eq. (15) is the *inverse* transform.

Other nomenclature of the Fourier transform

The Fourier transform is also represented as:

$$x(t) \overset{\mathcal{F}}{\Longleftrightarrow} X(j\Omega)$$

Other forms use $\mathcal{F}\{\}$ for the forward transforms and $\mathcal{F}^{-1}\{\}$ for the inverse transform:

$$X(j\Omega) = \mathcal{F}\{x(t)\}$$
$$x(t) = \mathcal{F}^{-1}\{X(j\Omega)\}$$

Alternate definitions

Definitions of the Fourier transform differ in other text and discipline. Some physics text adopt the *unitary* Fourier transform pair:

$$X(j\Omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$
$$x(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} X(j\Omega) e^{j\Omega t} d\Omega$$

Many engineering texts define the transform with respect to the frequency $F = 2\pi\Omega$.

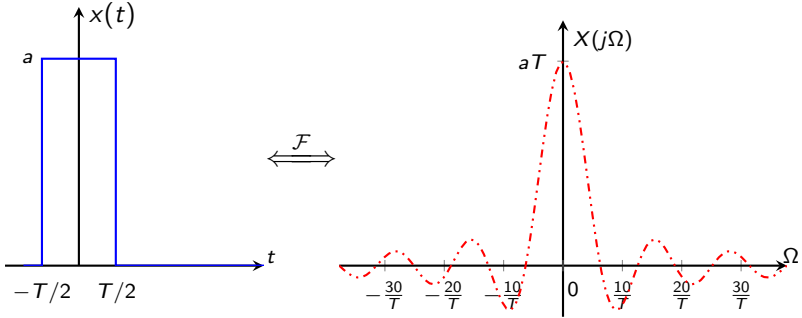
$$X(F) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi Ft} dt$$
$$x(t) = \int_{-\infty}^{\infty} X(F) e^{j2\pi Ft} dF$$

Some authors also drop the j from the notation and write $X(\Omega)$ or $X(F)$.

Fourier Transform Examples: Rectangular Pulse

Find the Fourier transform of the rectangular pulse function.

$$x(t) = \begin{cases} a, & |t| < T/2 \\ 0, & \text{otherwise} \end{cases}$$



Fourier Transform Examples: Rectangular Pulse (Solution)

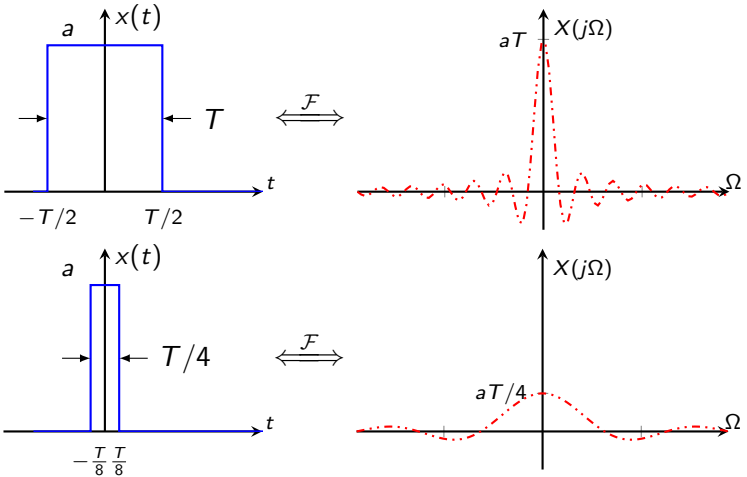
From the definition of the forward Fourier Transform

$$\begin{aligned} X(j\Omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt \\ &= a \int_{-\infty}^{\infty} e^{-j\Omega t} dt \\ &= a \left[\frac{j}{\Omega} e^{-j\Omega t} \right|_{-T/2}^{T/2} \\ &= \frac{ja}{\Omega} \left[e^{-j\Omega T/2} - e^{j\Omega T/2} \right] \\ &= aT \frac{\sin(\Omega T/2)}{\Omega T/2} \end{aligned}$$

The Fourier transform of a rectangular pulse is a sinc function ($(\sin x)/x$), centered around the $j\Omega = 0$ axis.

Fourier Transform of a Rectangular Pulse

As the pulse duration T decreases, the spectral bandwidth of the pulse (the main lobe of the sinc) increases.



Fourier transform of the Delta dirac function $\delta(t)$

Solution: The delta impulse function is defined as:

$$\delta(t) = \begin{cases} 0, & t \neq 0 \\ \text{undefined}, & t = 0. \end{cases}$$

And the "sifting" property as:

$$\int_{-\infty}^{\infty} \delta(t)dt = 1$$

Using the "sifting" property:

$$\int_{-\infty}^{\infty} \delta(t - T)x(t)dt = x(T)$$

When substituted in the forward Fourier transform

$$\Delta(j\Omega) = \int_{-\infty}^{\infty} \delta(t)e^{-j\Omega t}dt = 1 \quad \text{constant}$$

Fourier transform of a shifted impulse $\delta(t - T)$

Determine the Fourier transform of $\delta(t - T)$

$$F(j\Omega) = \int_{-\infty}^{\infty} \delta(t - T) e^{-j\Omega t} dt$$

Using the "sifting" property of the delta function

$$F(j\Omega) = \int_{-\infty}^{\infty} \delta(t - T) e^{-j\Omega t} dt = e^{-j\Omega T}$$

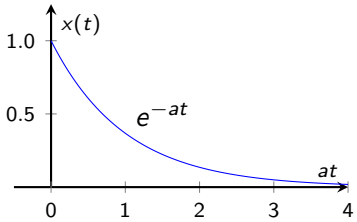
The Fourier transform of a shifted impulse is a **complex exponential**. It follows that the inverse Fourier transform of a complex exponential is an impulse, we have:

$$\mathcal{F}^{-1}\{e^{-j\Omega T}\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-j\Omega T} e^{j\Omega t} dt = \delta(T - t) = \delta(t - T)$$

Fourier Transform of an Exponential function

The one-sided real exponential function is defined as:

$$x(t) = \begin{cases} 0, & t < 0 \\ e^{-at}, & t \geq 0. \end{cases}$$



Fourier Transform of an Exponential function (Solution)

Solution: From the definition of the forward Fourier transform.

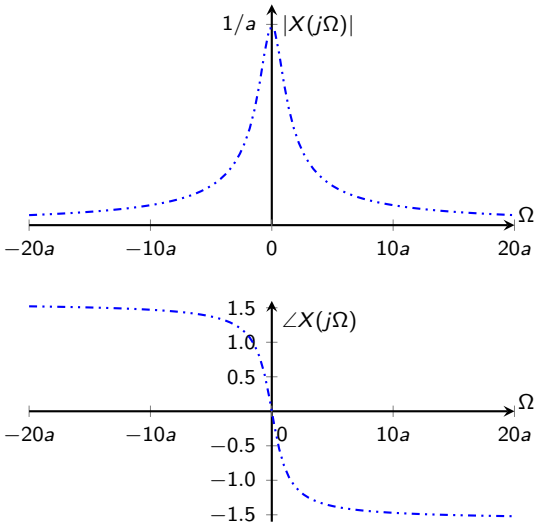
$$\begin{aligned} X(j\Omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt \\ &= \int_{0}^{\infty} e^{-at} e^{-j\Omega t} dt \\ &= \left[\frac{-1}{a + j\Omega} e^{-(a+j\Omega)t} \right|_{0}^{\infty} \\ &= \frac{1}{a + j\Omega} \end{aligned}$$

which is complex, in terms of magnitude and phase

$$|X(j\Omega)| = \frac{1}{\sqrt{a^2 + \Omega^2}}$$

$$\measuredangle X(j\Omega) = \tan^{-1}\left(\frac{-\Omega}{a}\right)$$

Magnitude and Phase plot: $X(j\Omega)$



Properties of the Fourier Transform

Existence of the Fourier transform and the Dirichlet conditions.

- The function $x(t)$ must be *intergrable in the absolute sense* over all time, that is

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

- There must be at most a finite number of maxima and minima in the function $x(t)$.
- There must be at most a finite number of discontinuities in the function $x(t)$ and all such discontinuities are finite in magnitude

Properties of the Fourier Transform

Linearity of the Fourier Transform

If

$$g(t) \overset{\mathcal{F}}{\Longleftrightarrow} G(j\Omega) \text{ and } h(t) \overset{\mathcal{F}}{\Longleftrightarrow} H(j\Omega)$$

Then for arbitrary constants a and b.

$$ag(t) + bh(t) \overset{\mathcal{F}}{\Longleftrightarrow} aG(j\Omega) + bH(j\Omega)$$

Properties of the Fourier transform

Duality.
If

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t}d\Omega \overset{\mathcal{F}}{\Longleftrightarrow} X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t}dt$$

then by definition

$$\mathcal{F}\{X(jt)\} = \int_{-\infty}^{\infty} X(jt)e^{-j\Omega t}dt = \int_{-\infty}^{\infty} X(jt)e^{j(-\Omega)t}dt = 2\pi x(-\Omega)$$

or

$$X(jt) \overset{\mathcal{F}}{\Longleftrightarrow} X(j\Omega) = 2\pi x(-\Omega)$$

where $X(jt)$ is $X(j\Omega)$ where Ω is replaced by t, and $x(-\Omega)$ is $x(t)$ where t is replaced with $-\Omega$.

Properties of the Fourier transform

Even and Odd functions. Let $x(t)$ be a real function written in terms of even and odd functions: $x(t) = x_e(t) + x_o(t)$, then the Fourier Transform is:

$$\begin{aligned} X(j\Omega) &= \int_{-\infty}^{\infty} (x_e(t) + x_o(t))(\cos(\Omega t) - j\sin(\Omega t))dt \\ &= \int_{-\infty}^{\infty} x_e(t)\cos(\Omega t)dt - j\int_{-\infty}^{\infty} x_o(t)\sin(\Omega t)dt \end{aligned}$$

- For real $x(t)$ the Fourier transform shows conjugate symmetry

$$X(-j\Omega) = X^*(j\Omega)$$

- Fourier transform of a real even function is a real even function
- Fourier transform of a real odd function is an imaginary odd function

Properties of the Fourier transform

Time Shifting. The Fourier transform of $x(t+\tau)$, time-shifted version of $x(t)$ is:

$$\mathcal{F}\{x(t+\tau)\} = e^{j\Omega\tau}X(j\Omega)$$

Time Scaling. If $x(t) \overset{\mathcal{F}}{\Longleftrightarrow} X(j\Omega)$, then

$$x(at) \overset{\mathcal{F}}{\Longleftrightarrow} \frac{1}{|a|}X\left(\frac{j\Omega}{a}\right)$$

where a is a constant, $a \neq 0$.
Time-Reversal. If $a = -1$, the time-saling property give:

$$x(-t) \overset{\mathcal{F}}{\Longleftrightarrow} X(-j\Omega)$$

Properties of the Fourier transform

Parseval's Theorem. The "energy" of an aperiodic function in the time domain is the integral of the "instantaneous power" over time.

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Parseval's theorem asserts the equivalence of energy in the time and frequency domain.

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\Omega)|^2 d\Omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega) X^*(j\Omega) d\Omega$$

The quantity $|X(j\Omega)|^2$ is a measure of energy of the function per unit bandwidth, and known as the **energy density** spectrum:

$$\Phi(j\Omega) = |X(j\Omega)|^2$$

Properties of the Fourier transform

Convolution. Given

$$x_1(t) \overset{\mathcal{F}}{\Longleftrightarrow} X_1(j\Omega) \text{ and } x_2(t) \overset{\mathcal{F}}{\Longleftrightarrow} X_2(j\Omega)$$

then,

$$x_1(t) * x_2(t) \overset{\mathcal{F}}{\Longleftrightarrow} X_1(j\Omega)X_2(j\Omega)$$

where the *convolution integral* is defined as:

$$x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)d\tau = \int_{-\infty}^{\infty} x_2(\tau)x_1(t-\tau)d\tau$$

Conversely, convolution in the frequency domain results in:

$$f(t)g(t) \overset{\mathcal{F}}{\Longleftrightarrow} \frac{1}{2\pi}F(j\Omega) * G(j\Omega)$$

Properties of the Fourier transform

Frequency shifting. Given $x(t) \overset{\mathcal{F}}{\Longleftrightarrow} X(j\Omega)$

$$x(t)e^{j\Omega_0 t} \overset{\mathcal{F}}{\Longleftrightarrow} X(j(\Omega - \Omega_0))$$

Time differentiation. If $x(t) \overset{\mathcal{F}}{\Longleftrightarrow} X(j\Omega)$, then

$$\frac{d}{dt}x(t) \overset{\mathcal{F}}{\Longleftrightarrow} (j\Omega)X(j\Omega)$$

and

$$\frac{d^n}{dt^n}x(t) \overset{\mathcal{F}}{\Longleftrightarrow} (j\Omega)^n X(j\Omega)$$

Properties of the Fourier transform

Integral of a function. Given $x(t) \overset{\mathcal{F}}{\Longleftrightarrow} X(j\Omega)$.

$$\mathcal{F}\Big\{ \int_{-\infty}^{t} x(\nu) d\nu \Big\} = \pi X(0)\delta(j\Omega) + \frac{1}{j\Omega} X(j\Omega)$$

where $X(0)$ is the average value of $x(t)$

$$X(0) = \int_{-\infty}^{\infty} x(t) dt$$

Examples: Fourier transform of a constant

Determine the Fourier transform of $x(t) = a$. The Fourier integral

$$X(j\Omega) = \int_{-\infty}^{\infty} ae^{-j\Omega t} dt$$

does not converge. But we know the Fourier transform of a rectangular pulse with duration T, is a $sin(x)/x$ function.

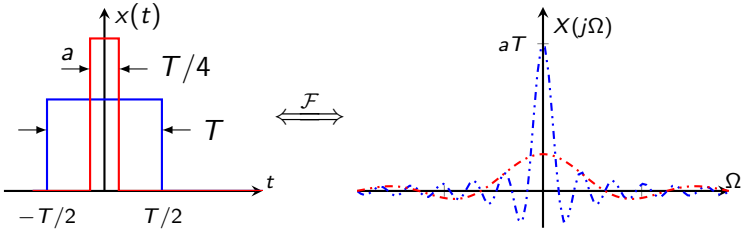
$$g(t) = \begin{cases} a, & |t| < T/2 \\ 0, & \text{otherwise} \end{cases}$$

$$G(j\Omega) = aT \frac{\sin(\Omega T/2)}{\Omega T/2}$$

As the duration T is increased, the peak amplitude if $G(j\Omega)$ increases and the bandwidth decreases.

Fourier Transform of a Rectangular Pulse

As the pulse duration T decreases, the spectral bandwidth of the pulse (the main lobe of the sinc) increases.



Since $x(t) = a = \lim_{T\to\infty} g(t)$, we have:

$$X(j\Omega) = \lim_{T\to\infty} G(j\Omega) = \lim_{T\to\infty} \left\{ aT \frac{\sin(\Omega T/2)}{\Omega T/2} \right\} = 2\pi a\delta(\Omega)$$

The Fourier transform of a constant is an impulse at $\Omega = 0$.

Example: Fourier transform of a unit-step

Determine the Fourier transform of $u(t)$:

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0. \end{cases}$$

The Fourier integral below, does not converge.

$$U(j\Omega) = \int_{-\infty}^{\infty} u(t) e^{-j\Omega t} dt = \int_{0}^{\infty} e^{-j\Omega t} dt$$

An alternative approach, start with the Fourier transform of the one-sided exponential function.

$$x(t) = \begin{cases} 0, & t < 0 \\ e^{-at}, & t \geq 0. \end{cases} \overset{\mathcal{F}}{\Longleftrightarrow} X(j\Omega) = \frac{1}{a + j\Omega}$$

Example: Fourier transform of a unit-step

Note that $u(t) = \lim_{a \to 0} x(t)$ implies that $U(j\Omega) = \lim_{a \to 0} X(j\Omega)$. Thus

$$X(j\Omega) = \frac{1}{a + j\Omega} = \frac{a}{a^2 + \Omega^2} - j\frac{\Omega}{a^2 + \Omega^2}$$

The real part is a Cauchy distribution, and for $a > 0$

$$\int_{-\infty}^{\infty} \frac{a}{a^2 + \Omega^2} d\Omega = \pi$$

and as $a \to 0$, $a/(a^2 + \Omega^2)$ becomes impulse-like, therefore as $a \to 0$.

$$\frac{a}{a^2 + \Omega^2} \to \pi\delta(\Omega) \text{ and } -j\frac{\Omega}{a^2 + \Omega^2} \to -j\frac{1}{\Omega}$$

The Fourier transform of the unit-step function is:

$$U(j\Omega) = \pi\delta(\Omega) + \frac{1}{j\Omega}$$

Fourier Transform of a Complex exponential

Determine the Fourier transform of $x(t) = e^{j\Omega_0 t}$, we have:

$$X(j\Omega) = \int_{-\infty}^{\infty} e^{j\Omega_0 t} e^{-j\Omega t} dt = \int_{-\infty}^{\infty} e^{-j(\Omega-\Omega_0)t} dt$$

It follows from the integral of the inverse Fourier transform of the complex exponential

$$\mathcal{F}^{-1}\{e^{-j\Omega T}\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-j\Omega(T-t)} dt = \delta(t - T)$$

The Fourier transform of a complex exponential is a shifted delta function in the frequency domain.

$$X(j\Omega) = \int_{-\infty}^{\infty} e^{-j(\Omega-\Omega_0)t} dt = 2\pi\delta(\Omega - \Omega_0)$$

Fourier Transform of a Sinusoid

Determine the Fourier transform of $x(t) = \cos(\Omega_0 t)$

$$\begin{aligned} X(j\Omega) &= \int_{-\infty}^{\infty} \cos(\Omega_0 t) e^{-j\Omega t} dt \\ &= \int_{-\infty}^{\infty} \Big[\frac{e^{j\Omega_0 t} + e^{-j\Omega_0 t}}{2}\Big] e^{-j\Omega t} dt \\ &= \frac{1}{2}\int_{-\infty}^{\infty} e^{j\Omega_0 t} e^{-j\Omega t} dt + \frac{1}{2}\int_{-\infty}^{\infty} e^{-j\Omega_0 t} e^{-j\Omega t} dt \\ &= \pi[\delta(\Omega - \Omega_0) + \delta(\Omega + \Omega_0)] \end{aligned}$$

The Fourier transform is purely real.
It can be shown that the Fourier transform of a $\sin(\Omega_0 t)$ is purely imaginary as:

$$\mathcal{F}\{\sin(\Omega_0 t)\} = -j\pi[\delta(\Omega - \Omega_0) - \delta(\Omega + \Omega_0)]$$

Example: Frequency shift property

Given $g_1(t) = 2\cos(200\pi t)$ and $g_2(t) = 5\cos(1000\pi t)$.
Find the Fourier transform of

$$g_3(t) = g_1(t)g_2(t)$$

Answer: $G_3(j\Omega) = 5\pi[\delta(\Omega - 1200\pi) + \delta(\Omega + 1200\pi)] + 5\pi[\delta(\Omega - 800\pi) + \delta(\Omega + 800\pi)]$

Table of Fourier Transforms

Table: Table of Fourier Transforms

	$f(t)$	$F(\omega)$	
1	$e^{-at}u(t)$	$\frac{1}{a+j\omega}$	$a > 0$
2	$e^{at}u(-t)$	$\frac{1}{a-j\omega}$	$a > 0$
3	$e^{-a\|t\|}$	$\frac{2a}{a^2+\omega^2}$	$a > 0$
4	$te^{-at}u(t)$	$\frac{1}{(a+j\omega)^2}$	$a > 0$
5	$t^n e^{-at}u(t)$	$\frac{n!}{(a+j\omega)^{n+1}}$	$a > 0$
6	$\delta(t)$	1	
7	1	$2\pi\delta(\omega)$	
8	$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$	

Table of Fourier Transforms (continued)

Table: Table of Fourier Transforms (continued)

	$f(t)$	$F(\omega)$	
9	$\cos(\omega_0 t)$	$\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$	
10	$\sin(\omega_0 t)$	$\frac{\pi}{j}[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$	
11	$u(t)$	$\pi\delta(\omega)+\frac{1}{j\omega}$	
12	sgn t	$\frac{2}{j\omega}$	
13	$\cos(\omega_0 t)u(t)$	$\frac{\pi}{2}[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]+\frac{j\omega}{\omega_0^2-\omega^2}$	
14	$\sin(\omega_0 t)u(t)$	$\frac{\pi}{2j}[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]+\frac{\omega_0}{\omega_0^2-\omega^2}$	
15	$e^{-at}\sin(\omega_0 t)u(t)$	$\frac{\omega_0}{(a+j\omega)^2+\omega_0^2}$	$a>0$
16	$e^{-at}\cos(\omega_0 t)u(t)$	$\frac{a+j\omega_0}{(a+j\omega)^2+\omega_0^2}$	$a>0$

Table of Fourier Transforms (continued)

Table: Table of Fourier Transforms [1]

	$f(t)$	$F(\omega)$	
17	$\text{rect}(\frac{t}{\tau})$	$\tau\text{sinc}(\frac{\omega\tau}{2})$	
18	$\frac{W}{\pi}\text{sinc}(Wt)$	$\text{rect}(\frac{\omega}{2W})$	
19	$\Delta(\frac{t}{\tau})$	$\frac{\tau}{2}\text{sinc}^2(\frac{\omega\tau}{4})$	
20	$\frac{W}{2\pi}\text{sinc}^2(\frac{Wt}{2})$	$\Delta(\frac{\omega}{2W})$	
21	$\sum_{n=-\infty}^{\infty}\delta(t-nT)$	$\omega_0\sum_{n=-\infty}^{\infty}\delta(\omega-n\omega_0)$	$\omega_0=\frac{2\pi}{\tau}$
22	$e^{-t^2/2\sigma^2}$	$\sigma\sqrt{2\pi}e^{-\sigma^2\omega^2/2}$	

[1] Signal Processing and Linear Systems, B.P. Lathi, Berkeley-Cambridge Press, 1998

Table of Fourier Transforms Properties

Table: Table of Fourier Transforms [2]

Operation	$f(t)$	$F(\omega)$
Linearity	$af_1(t) + bf_2(t)$	$aF_1(\omega) + bF_2(\omega)$
Duality / Symmetry	$F(t)$	$2\pi f(-\omega)$
Scaling (a is real)	$f(at)$	$\frac{1}{\lvert a\rvert} F\left(\frac{\omega}{a}\right)$
Time-shift	$f(t - t_0)$	$F(\omega)e^{-j\omega t_0}$
Frequency-shift (ω_0 is real)	$f(t)e^{j\omega t}$	$F(\omega - \omega_0)$
Time convolution	$f_1(t) * f_2(t)$	$F_1(\omega)F_2(\omega)$

[2] Signal Processing and Linear Systems, B.P. Lathi, Berkeley-Cambridge Press, 1998

Table of Fourier Transforms Properties (continued)

Table: Table of Fourier Transforms [3]

Operation	$f(t)$	$F(\omega)$
Frequency convolution	$f_1(t)f_2(t)$	$\frac{1}{2\pi}F_1(\omega) * F_2(\omega)$
Time differentiation	$\frac{d^n f}{dt^n}$	$(j\omega)^n F(\omega)$
Time integration	$\int_{-\infty}^{t} f(x)dx$	$\frac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega)$

[3] Signal Processing and Linear Systems, B.P. Lathi, Berkeley-Cambridge Press, 1998